CHAPTER 1

INTRODUCTION

1.1 Overview

The development of advanced prosthetic systems has become increasingly important as the demand for affordable, intuitive, and versatile assistive technologies continues to rise. Traditional prosthetic arms often rely on a single mode of control—such as mechanical switches or basic electromyography limiting their adaptability and ease of use for individuals with diverse physical abilities. To overcome these limitations, modern prosthetics must integrate multiple sensing technologies, intelligent processing, and user-friendly interfaces to better replicate natural human arm functions. The AI-Powered Multi-Mode Controlled Prosthetic Arm introduces a novel solution by combining embedded microcontrollers, bio signal processing, artificial intelligence, and multi-modal human machine interaction. At the core of the system, an Arduino Nano serves as the primary control unit, while an Arduino Uno handles complex EEG/EMG signal processing for mind-controlled actuation. The addition of an ESP32-CAM enhances the prosthetic's intelligence by enabling real-time computer vision, object detection, and environment monitoring. To ensure flexible and intuitive control, the system incorporates multiple input channels, including voice recognition, gesture detection using ultrasonic sensors, motion tracking via gyroscope data, and neuro-muscular signals. These diverse input modes allow the prosthetic to adapt to the user's preferences, physical condition, and situational needs. A robust power management system featuring voltage regulators, UBEC modules, a stable 12V supply, and a power distribution board ensures reliable operation of high-current servos and microcontrollers. Through the integration of hardware intelligence and AI-assisted perception, this prosthetic arm aims to provide a more natural, responsive, and autonomous user experience. The system demonstrates a promising step toward low-cost, high-performance assistive technology that bridges the gap between human intention and robotic execution.

1.2 PROBLEM STATEMENT

Many individuals who have lost their arm struggle with traditional prosthetic limbs due to limited control, unnatural movements, and lack of responsiveness. These devices often require manual operation or basic muscle signals, which can be slow, inaccurate, and frustrating for users. As a result, performing everyday tasks becomes challenging and reduces the user's quality of life. There is a growing need for an advanced prosthetic solution that can interpret brain or muscle signals more effectively and provide smooth, real-time control. By integrating artificial intelligence with mind-controlled technology, we can create a smarter, more intuitive prosthetic arm that feels like a natural extension of the body.

1.3 OBJECTIVES

1.Multimodal Control Integration

- To implement multiple input modes EMG, EEG, voice, gesture, motion, and vision to allow flexible user interaction based on physical ability and environment.
- To design a fusion framework that prioritizes the most accurate and stable control modality in real time.

2. Bio signal Acquisition & Processing

- To acquire and process EMG and EEG signals using Arduino Uno for intent recognition.
- To develop reliable filtering, feature extraction, and classification methods for accurate detection of user intention.

3. Vision-Based Assistance

- To utilize ESP32-CAM for object recognition, grasp type suggestion, or autonomous control support.
- To implement lightweight vision algorithms suitable for embedded hardware limitations.

5. Gesture and Motion Sensing

- To incorporate ultrasonic gesture sensing and gyroscope/accelerometer data for hand or arm movement control.
- To apply sensor-fusion algorithms (Kalman/complementary filters) for stable motion interpretation.

6. Voice-Control Functionality

- To use VR3 module for implementing offline voice command recognition for high-level control and mode switching.

1.4 LITERATURE SURVEY

A literature survey is a critical and systematic review of existing research, studies, and technical papers related to the chosen project topic. It helps in understanding the current state of technology, previously proposed methods, and ongoing advancements in the field. By studying earlier works, the strengths, limitations, and research gaps in existing systems can be clearly identified. In the context of our project on an AI-powered multimode controlled prosthetic arm, the literature survey focuses on various control techniques such as EEG-based brain–computer interfaces, gesture recognition systems, sensor-based control, and intelligent optimization algorithms. It also reviews different hardware architectures, signal processing techniques, and machine learning approaches used in prosthetic and rehabilitation devices.

[1] "Prosthetic AI Enabled Arm for Rehabilitation and Advanced Dynamics"

Existing System

In paper titled "Prosthetic AI Enabled Arm for Rehabilitation and Advanced Dynamics" by Mahantesh K et al. presents a Brain-Computer Interface (BCI)-based prosthetic arm system that utilizes motor imagery EEG signals to control limb movement. EEG data is collected non-invasively from participants performing or imagining specific arm and hand gestures, which is then pre-processed using techniques like PCA and ICA to remove noise and artifacts. Features are extracted and classified using machine learning algorithms including SVM, KNN, Naive Bayes, Logistic Regression, and Random Forest—of which Random Forest yielded the highest accuracy. The classified signals are transmitted via Bluetooth (ESP32) to an Arduino, which controls a motorized prosthetic hand capable of executing various tasks such as grasping, wrist rotation, and finger movements. Simulated through Tinker CAD, the system aims to restore mobility and functionality for upper limb amputees. The study demonstrates the feasibility and effectiveness of combining EEG-based BCI with AI to develop a responsive and adaptive prosthetic arm, with future enhancements suggested through deep learning and expanded motion capabilities.

Proposed system:

The proposed system introduces an AI-powered multimode controlled prosthetic arm designed to enhance rehabilitation and functional performance. The system integrates multiple input methods such as gesture control, voice commands, and EEG-based mind control, enabling flexible and user-friendly operation. Artificial intelligence techniques are used to interpret sensor data and generate accurate control signals for smooth and natural arm movements. A 3D printed prosthetic arm is used to achieve a lightweight, cost-effective, and customizable design. The proposed system improves motion accuracy, adaptability, and user comfort while supporting real-time response and intelligent control, making it suitable for both rehabilitation and daily assistance.

[2] "Smart Brain controlled prosthetic arm using hybrid Binary PSO"

Existing system:

In paper "Smart Brain controlled prosthetic arm using hybrid Binary PSO" by Dr. K. Suresh, et al. presents a low-cost, non-invasive brain-controlled prosthetic arm using EEG signals. It uses a hybrid Binary Particle Swarm Optimization (PSO) with Genetic Algorithm (GA) for feature extraction and classification of brain signals, primarily based on eye blinks and attention levels. These signals are processed in MATLAB and sent to an Arduino to control a 3D-printed robotic arm with movements like elbow flexion and finger gripping. The system was tested on three users and showed reliable performance, offering a simple and affordable solution for prosthetic limb control. A general AI-powered multimode controlled prosthetic arm uses advanced technology like deep learning and high-quality brain signals to control the arm with smooth, natural movements. It can handle complex tasks and is often more accurate and responsive.

Proposed system:

The proposed system in our project overcomes these limitations by implementing an AI-powered multimode controlled prosthetic arm. While brain signals remain an important control input, the system integrates gesture control, voice commands, and EEG-based mind control to ensure reliable operation under different conditions. Artificial intelligence techniques are used to process sensor data and improve decision-making accuracy instead of relying solely on Hybrid Binary PSO. A 3D-printed prosthetic arm is employed to achieve a lightweight, low-cost, and customizable design. The proposed system provides smoother motion, faster response, improved accuracy, and greater user comfort, making it suitable for real-time use and rehabilitation.

[3] "AI Enables Real-Time and Intuitive Control of Prostheses via Nerve Interface"

Existing system:

In paper “AI Enables Real-Time and Intuitive Control of Prostheses via Nerve Interface” by Zhi Yang, Edward W. Keefer, et al. presents a neuroprosthetic system that uses artificial intelligence to decode movement intentions from peripheral nerve signals in amputees. By employing a recurrent neural network (RNN), the system interprets multichannel nerve data to control a prosthetic hand with six degrees of freedom, including individual finger and wrist movements. The AI agent achieves high accuracy (up to 98%) and operates in real-time, with a reaction time of approximately 0.8 seconds and an information throughput of 6.09 bits per second. Tested over a 16-month period, the system proved stable, intuitive, and reliable, offering a significant advancement toward natural, thought-controlled prosthetic limbs.

Proposed system:

A typical AI-powered multimode controlled prosthetic arm uses brain signals, like EEG, to interpret a user's thoughts and control a robotic arm, often focusing on general mental commands such as concentration or eye blinks. These systems are usually non-invasive but have limited accuracy and control due to noisy signals and fewer input channels. In contrast, the system in this paper uses a more advanced approach by decoding motor intent directly from peripheral nerve signals using implanted electrodes. This method captures detailed, high-quality data about intended finger and wrist movements. The system then uses a recurrent neural network (RNN) to provide real-time, intuitive, and highly accurate control over a prosthetic hand, achieving up to 98% accuracy and maintaining long-term performance over 16 months. It allows for fine motor control and natural movement, far beyond what is possible with basic brainwave-based systems.

[4] “Design and Build of bionic Arm”

Existing system:

In paper “Design and Build of a Bionic Arm” by Rami Amin, et al. presents a low-cost, 3D-printed prosthetic arm that is controlled using a wearable sensor suit capable of detecting hand gestures. It aims to offer an accessible solution for limb replacement, suitable for both civilian and military applications. The researchers are also working on integrating advanced algorithms to improve motion control and task complexity handling. The paper “Design and Build of a Bionic Arm” focuses on building a gesture-controlled prosthetic arm using 3D printing and wearable sensor suits that detect arm/hand movements. It’s a practical, cost-effective solution aimed at improving accessibility.

Proposed system:

The proposed system in our project focuses on an AI-powered multimode controlled prosthetic (bionic) arm that improves functionality, flexibility, and user comfort. The system integrates multiple control methods such as gesture control, voice commands, and EEG-based mind control, allowing intuitive operation based on the user's ability. Artificial intelligence techniques are used to process sensor data and generate accurate control signals. A 3D-printed prosthetic arm is used to reduce cost and weight while enabling customization. The proposed system provides smoother, more natural movements, real-time response, and improved adaptability, making it suitable for both rehabilitation and practical daily applications.

[5] "Controlling a Below the-Elbow Prosthetic arm using foot controller"

Existing system:

In paper "Controlling a Below the-Elbow Prosthetic arm using foot controller" by Jack Wilgus, et al. presents a novel approach to prosthetic limb control by utilizing foot movements instead of traditional methods like myoelectric signals. The system consists of an insole controller and a sensor-controller unit attached to the shoe or sock, which detects various foot movements—such as dorsiflexion, plantarflexion, inversion, and eversion—to control the wrist functions of a 3D-printed prosthetic arm called the Infinity Arm. Additionally, toe push buttons are used to execute hand grips and gestures. Commands are transmitted wirelessly, allowing real-time control. The prosthetic arm features five fingers capable of gripping lightweight objects, with up to 120° wrist rotation and 70° of flexion/extension. It also includes haptic feedback through fingertip force sensors that send vibrations to an armband, giving the user a sense of touch. This method offers an intuitive, cost-effective, and muscle-fatigue-free alternative to conventional prosthetic controls, making it accessible and practical for daily use.

Proposed system:

The proposed system in our project introduces an AI-powered multimode controlled prosthetic arm that eliminates dependency on foot-based control. Instead, the prosthetic arm is operated using gesture control, voice commands, and EEG-based mind control, providing more natural and intuitive interaction. AI algorithms process sensor data in real time to generate accurate and smooth arm movements. The use of a 3D-printed prosthetic arm ensures lightweight, cost-effective, and customizable design. The proposed system improves comfort, reduces physical strain, and enhances independence, making it more suitable for daily activities and rehabilitation

1.5 EXISTING AND PROPOSED SYSTEM SUMMARY

The current prosthetic arm systems available in the market primarily rely on mechanical switches, body-powered mechanisms, or surface electromyography (EMG) signals for operation. While these systems have evolved over time, they still face significant limitations in terms of precision, adaptability, and ease of use. Most conventional prosthetics require the user to learn specific movements or muscle contractions to trigger certain actions, which can be physically and mentally exhausting. Furthermore, the response time is often slow, and the movements lack fluidity and natural coordination. These systems generally do not adapt to the individual user's changing muscle patterns or intentions over time. As a result, users often experience frustration, discomfort, and difficulty in performing everyday tasks, such as eating, writing, or dressing. The absence of intelligent feedback and real-time adaptability in existing systems limits their effectiveness and reduces user satisfaction. There is a clear need for more advanced technology that bridges the gap between intention and action, offering a prosthetic solution that is both intuitive and responsive.

The proposed system introduces an advanced AI-powered, multi-mode controlled prosthetic arm designed to provide users with highly intuitive, flexible, and reliable control over artificial limb movements. Unlike traditional prosthetic devices that rely on a single input method, this system integrates multiple control channels—such as voice commands, hand gestures, IMU-based motion sensing, EMG muscle signals, and optional EEG-based intent detection—to ensure that the arm can adapt to different user environments and physical conditions. By combining these diverse inputs, the system aims to offer a more natural interaction experience and overcome the limitations posed by fatigue, noise, or sensor failure in single-mode systems. At the core of the prosthetic arm is a hybrid processing architecture consisting of a high-performance microcontroller and an AI-enabled processor. Sensors such as ultrasonic modules, IMUs, cameras, microphones, and EMG electrodes continuously capture data that represent the user's intention or the surrounding environment. This data is processed through a set of machine-learning algorithms that perform gesture classification, voice recognition, object detection, and signal fusion to interpret the user's desired action. The system uses intelligent decision-making logic to merge these inputs and generate the most accurate control command for the prosthetic arm, thereby enhancing precision and reducing response delays. To convert interpreted intentions into physical actions, the prosthetic arm employs a set of servo or BLDC motors controlled through PWM signals generated by the microcontroller. These motors manage finger movement, wrist rotation, and grip force based on the commands received from the AI controller. Additional force and position feedback sensors are integrated into the mechanical structure to ensure safe operation, preventing accidental crushing or overextension during use. A dedicated battery and voltage regulation system supply stable power to all components, with thermal and overload protection for operational safety.

1.6 Hardware and Software Components used

HARDWARE COMPONENTS

- Arduino Nano
- Arduino Uno
- Servo motors
- LCD display
- ESP32-CAM
- Voltage Regulator
- Heatsink
- Power Supply
- Voice Recognition Module
- Ultrasonic Sensor
- Gyroscope
- EEG Sensor
- 3D Printed Prosthetic Arm

SOFTWARE COMPONENTS

- Arduino IDE
- LIBRARIES
    - SoftwareSerial.h
    - VoiceRecognitionV3.h
    - Servo.h
    - LiquidCrystal.h
    - Wire.h

1.7 Organization of Report

The project report organization is as follows:

Chapter 2: “System design and implementation” describes about the system by using Block diagram and explains about typical operations of the system.

Chapter 3: “Hardware Design and Description” lists various hardware components, their description and the features, implemented in the system.

Chapter 4: “Hardware Implementation” gives the description about the circuit diagrams used for AI Powered Multimode Controlled Prosthetic Arm and describes the working of the system.

Chapter 5: “Software Description” gives a brief description about the software components used such as Arduino IDE.

Chapter 6: “Software Implementation” describes flowchart and algorithm for AI Powered Multimode Controlled Prosthetic Arm that describes software part of the implementation of the system.

Chapter 7: “Experimental results” describes the various results and their description when corresponding intended operations were performed. The results of all the test cases have been described along with the snapshots of each outcome.

Chapter 8: This chapter describes about Advantages, Disadvantages and Applications of the implemented system.

Chapter 9: This chapter describes about Conclusion and Future Scope of the implemented system.

1.8 Summary

This chapter provides a brief overview by describing the challenges faced by physically challenged individuals in performing daily activities and explains how these challenges are addressed through the proposed AI Powered Multimode Controlled Prosthetic Arm. The chapter also explains the implementation of the system using multiple control modes such as gesture control, voice commands, and EEG-based mind control. The drawbacks of existing prosthetic systems are discussed, and the manner in which the proposed system overcomes these limitations is highlighted. This chapter also provides a brief overview of the organization of the report, giving an outline of the upcoming chapters.

CHAPTER 2

SYSTEM DESIGN AND OPERATION

This chapter presents the detailed design and description of the proposed AI Powered Multimode Controlled Prosthetic Arm. It explains the overall system architecture, functional blocks, and the interaction between hardware and software components. The system is designed to provide intuitive, accurate, and real-time control of a prosthetic arm using multiple input modes such as gesture control, voice commands, and EEG-based mind control. Emphasis is given to modular design, reliability, and user comfort, ensuring efficient signal acquisition, processing, and actuation. The chapter also describes how the collected sensor data is processed by the microcontroller and converted into precise movements of the 3D-printed prosthetic arm.

2.1 BLOCK DIAGRAM

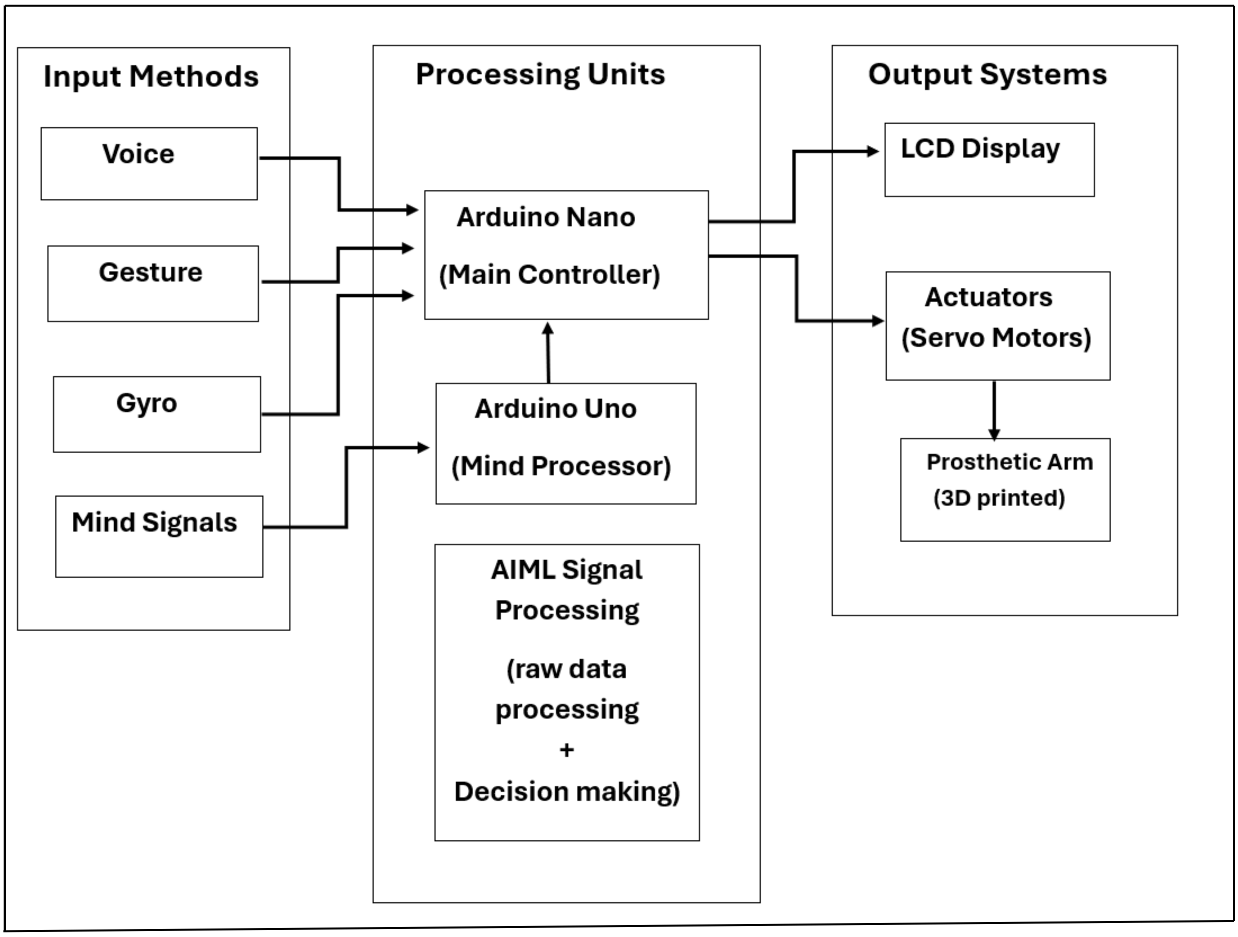


Figure 2.1 Block diagram of AI powered multimode controlled prosthetic arm

2.2 Working Mechanism

The given block diagram illustrates the working of an AI-powered multimode controlled prosthetic arm system, which is designed to help amputees control an artificial arm using multiple input methods such as voice, gesture, gyro, and mind signals as shown in the figure2.1. The system integrates embedded controllers, AI/ML-based signal processing, and electromechanical actuators to achieve smooth and accurate arm movements.

The working begins with the input methods, where different sensors capture user intent. The voice module detects spoken commands and converts them into digital signals. The gesture recognition unit identifies hand movements or postures, while the gyroscope sensor measures orientation and angular motion to support stable and accurate arm positioning. Simultaneously, mind signals such as EEG or EMG are collected to represent neural or muscle activity associated with intended movements. These raw signals are sent to the processing units. The Arduino Uno, acting as the mind processor, primarily handles EEG/EMG data. It filters noise, amplifies relevant signals, and extracts meaningful features that represent user intention. The processed mind-signal data is then forwarded to the Arduino Nano, which serves as the main controller of the system.

An AI/ML signal processing module operates alongside the controllers to analyze inputs from all modes. This module performs raw data processing, pattern recognition, and decision-making. Based on predefined models or trained algorithms, it selects the appropriate control mode or fuses multiple inputs to generate an accurate command for the prosthetic arm. The Arduino Nano, as the central control unit, integrates decisions from the AI/ML module and the Arduino Uno. It translates high-level commands (such as grip, release, or rotate) into low-level control signals suitable for actuators. At the same time, system status and selected modes are displayed on the LCD display, providing real-time feedback to the user. Finally, the output system executes the commands. The servo motors act as actuators, converting electrical control signals into mechanical motion. These servos drive the joints of the 3D-printed prosthetic arm, enabling smooth and controlled movements that mimic natural arm actions.

CHAPTER 3

HARDWARE DESIGN AND DESCRIPTION

This chapter describes the hardware design and components used in the proposed AI Powered Multimode Controlled Prosthetic Arm. The hardware system is designed to ensure reliable sensing, efficient processing, and precise actuation of the prosthetic arm. It consists of various sensors, control units, actuators, and power supply modules that work together to achieve real-time and accurate operation. The selection of each hardware component is based on performance, compatibility, cost-effectiveness, and ease of integration. This chapter explains the role, specifications, and interconnection of all hardware components involved in the implementation of the system.

3.1Arduino NANO

The Arduino Nano is a compact microcontroller board that functions as the central processing unit of the system. It operates by receiving various input signals from sensors, converting them into digital form using its built-in analog-to-digital converter, and processing them according to the programmed logic as shown in the figure 3.1. After interpreting the sensor data, the Arduino makes real-time decisions and generates appropriate output signals to control actuators such as servo motors, DC motors, or displays. Through pulse-width modulation and communication interfaces, it ensures precise motor control and smooth interaction with external modules. Thus, the Nano works as a small embedded computer that coordinates input sensing, signal processing, decision-making, and output control within an electronic system.

Figure 3.1 Arduino Nano

The Arduino Nano acts as the central microcontroller that coordinates different sensing and control modes in an AI-powered multimode prosthetic arm. It receives input signals from various sensors such as EMG electrodes, gyroscope, voice module and converts these signals into digital data. The AI model sends classification or mode-selection commands to the Arduino

Nano, which then processes these commands using its built-in ATmega328P microcontroller. Based on the selected mode, the Nano generates precise PWM control signals to drive servo motors responsible for finger, wrist, or elbow movement. It ensures smooth motion by controlling speed, angle, and torque while managing power distribution to the actuators. Overall, the Arduino Nano acts as the bridge between AI decision-making and the mechanical movement of the prosthetic arm, ensuring fast, reliable, and accurate control of the artificial limb.

3.2Arduino UNO

The Arduino Uno is a widely used microcontroller board based on the ATmega328P, and it serves as a central controller in electronic systems. It operates by reading inputs from various sensors, converting analog signals into digital values through its built-in ADC, and processing these signals using preprogrammed instructions as shown in the above figure3.2. The Uno analyzes the incoming data, performs logical operations, and makes decisions according to the application requirements. After processing, it sends output signals to actuators such as servo motors, DC motors, displays, or communication modules. Using its PWM pins, it controls motor speed and position with high precision, while its digital pins handle switching and signaling tasks. Through serial, I²C, and SPI communication, the Arduino Uno can also interact with external sensors, wireless modules, and AI components. Overall, it works as a reliable embedded controller capable of sensing, processing, and executing tasks efficiently.

Figure 3.2 Arduino Uno

The Arduino UNO functions as the primary microcontroller that links the user's input signals with the mechanical actions of an AI-powered multimode prosthetic arm. It collects data from different sensors such as EEG electrodes, voice recognition modules, and gyroscopes, converting their analog to digital signals into meaningful control inputs. These sensor readings are either processed directly on the UNO or sent to an external AI module, which decides the user's intended action and selects the appropriate operating mode. Once the AI output is received, the Arduino UNO generates precise PWM signals to activate servo motors or motor drivers, producing controlled finger, wrist, or elbow movement. It also handles feedback elements like LEDs, buzzers, or small displays to indicate the active mode. Overall, the Arduino UNO acts as a real-time coordinator that interprets sensor data, executes AI-based commands, and ensures smooth, safe, and accurate motion of the prosthetic arm.

3.3 ESP32-CAM

The ESP-32 CAM is a compact microcontroller module that integrates a powerful dual-core ESP32 processor with a built-in camera and Wi-Fi/Bluetooth communication features as shown in figure 3.3. It works by capturing images or video through the OV2640 camera sensor and processing them using the onboard microcontroller. The ESP-32 CAM can run embedded image-processing algorithms, face detection, object tracking, or lightweight machine-learning models. After processing the visual data, it can send results wirelessly through Wi-Fi or Bluetooth to a central controller, cloud server, or mobile device. The module supports real-time streaming, remote monitoring, and IoT connectivity, making it suitable for vision-based automation and smart devices. Its built-in GPIO pins also allow interfacing with external sensors and actuators, enabling image-based decision-making within embedded systems.

Figure3.3 ESP32-CAM

The ESP32-CAM serves as the vision-processing and wireless communication module in an AI-powered multimode prosthetic arm, enabling camera-based control and intelligent mode switching. Its built-in camera captures real-time images of the hand gestures, which are processed either on the ESP32-CAM using lightweight AI models for advanced image recognition. Based on this visual analysis, the system can identify objects, detect gestures, recognize user-defined commands, or automatically select grip patterns such as power grip, or precision hold. The ESP32-CAM then communicates the interpreted command to the primary microcontroller (Arduino/ESP32 main board), which drives the servo motors accordingly to produce the required limb movement. Overall, the ESP32-CAM adds AI-vision capability, enhances multimodal control, and allows the prosthetic arm to respond intelligently to both user gestures and environmental cues.

3.4 Voltage Regulator (LM7085)

A voltage regulator is an electronic device or circuit used to maintain a constant and stable output voltage regardless of variations in input voltage, load current, or environmental conditions. It ensures that sensitive electronic components receive the exact voltage they require for proper operation as shown in the above figure3.4. Voltage regulators can be linear (such as the 7805 series) or switching types (such as buck and boost converters). Linear regulators provide a smooth, noise-free output but are less efficient, while switching regulators offer higher efficiency by rapidly switching the power transistor on and off. In operation, the regulator compares the output voltage with a reference voltage and adjusts the internal circuitry to keep the voltage steady even when the battery drains or the load changes. This stable supply is essential for microcontrollers, sensors, communication modules, and motors that need precise and reliable power. Since the output from the piezo sensor can vary with the amount of pressure.

Figure 3.4 voltage regulator (LM7085)

The voltage regulator in an AI-powered multimode prosthetic arm ensures that all electronic components such as microcontrollers, sensors, wireless modules, and servo motors receive a stable and safe supply voltage regardless of battery fluctuations. Since the prosthetic arm is powered by rechargeable Li-ion batteries whose voltage changes as they discharge, the regulator converts this varying input into a constant output. This protects sensitive components like the Arduino, ESP32-CAM, IMU, EMG amplifier, and voice module from damage due to overvoltage while preventing malfunction caused by undervoltage. It also isolates high-current motor power from low-current logic circuits, ensuring smooth and noise-free operation of sensors during motor movement.

3.5 Heatsink

A heat sink is a passive cooling device designed to absorb and dissipate excess heat generated by electronic components during operation as shown in the above figure3.5. It is typically made from aluminum or copper due to their high thermal conductivity. The heat sink increases the surface area exposed to air, allowing heat to be transferred away from sensitive components such as voltage regulators, microprocessors, motor drivers, and power transistors. As the electronic part heats up, the heat is conducted into the heat sink and then released into the surrounding air through convection. Fins or ridges on the heat sink further improve heat dissipation by maximizing airflow contact. This prevents overheating, thermal damage, and performance degradation, ensuring reliable and safe operation of electronic circuits.

Figure3.5 Heat sink

The heatsink in an AI-powered multimode prosthetic arm is used to dissipate excess heat generated by components such as voltage regulators, motor drivers, microcontrollers, and high-current servo motors. During continuous operation, these electronic parts can heat up due to power losses and electrical load, which may lead to reduced performance or permanent damage. The heatsink absorbs this heat through direct contact with the component and spreads it over a larger surface area, allowing faster cooling through natural air convection. In a prosthetic arm, where compact design and safety are essential, the heatsink ensures that temperature-sensitive devices like the motor driver (e.g., L298, TB6612), power regulators, and

AI processing modules remain within safe operating limits. By maintaining stable thermal conditions, the heatsink improves efficiency, prolongs component life, and ensures reliable, smooth, and safe operation of the prosthetic arm even under heavy movement or multi-mode usage.

3.6 Voice recognition module

A voice recognition module is an electronic device that identifies and processes human voice commands and converts them into digital signals that a microcontroller can understand as shown in the above figure3.6. It uses a microphone to capture the spoken input and then applies signal processing techniques such as noise filtering, feature extraction, and pattern matching to recognize specific words or phrases. Many modules, such as the Elechouse Voice Recognition Module are capable of storing predefined voice commands and comparing incoming speech with the trained database. Once a command is recognized, the module sends a corresponding digital output or serial signal to the controller, enabling hands-free operation. These modules enhance user interaction by allowing voice-based control in embedded and automation systems.

Figure3.6 Voice Recognition module

In an AI-powered multimode prosthetic arm allows the user to control the arm through spoken commands, providing a hands-free and intuitive control mode. The module contains a built-in microphone and a small processor that captures the user’s voice, filters background noise, and converts the spoken words into digital command codes. These recognized commands such as “open hand,” “close hand,” “rotate wrist,” or “change mode” are sent to the main microcontroller (Arduino, ESP32) which interprets them and activates the appropriate servo motors or control functions. AI integration improves accuracy by enabling keyword detection, custom command training, and adaptive noise filtering, allowing the prosthetic arm to respond reliably even in noisy environments. Overall, the voice recognition module acts as a natural language interface, enabling easy mode switching and movement control, enhancing accessibility, and making prosthetic operation more user-friendly.

3.7 Servo motor

A servo motor is a precision-controlled actuator that provides accurate angular or linear positioning based on input signals. It typically consists of a small DC motor, a gear mechanism, a feedback potentiometer, and a control circuit enclosed in a compact unit as shown in the above figure3.7. The servo receives a control signal in the form of Pulse Width Modulation (PWM), where the width of the pulse determines the exact angle to which the motor shaft should rotate, usually within a range of 0° to 180°. The internal feedback system continuously compares the actual position of the shaft with the desired position and makes real-time corrections, ensuring smooth, stable, and accurate movement. Because of their high torque, fast response, and precise controllability, servo motors are widely used in robotics, automation, and mechatronics.

Figure3.7 Servo Motor

In an AI-powered multimode controlled prosthetic arm, the servo motor plays a crucial role as the primary actuator responsible for performing precise and controlled movements such as finger bending, wrist rotation, and elbow lifting. A servo motor is a closed-loop electromechanical device that can rotate to a specific angle based on the command it receives from the controller (Arduino, or AI-processing unit).

3.8 LCD display:

An LCD (Liquid Crystal Display) is a flat, low-power electronic display used to visually present information in embedded systems as shown in the above figure3.8. It works by controlling the alignment of liquid crystal molecules between transparent electrodes, which either block or allow light to pass through when an electric field is applied. Common types like the 16×2 or 20×4 LCD modules display characters, numbers, and symbols, while some use an LED backlight for better visibility. The microcontroller sends digital data and control signals to the LCD through data pins, enabling it to display messages, system status, sensor readings, or mode information. Due to its low power consumption, compact size, and ability to display real-time information clearly, LCDs are widely used in electronic devices, measurement instruments, and automation systems.

Figure3.8 LCD Display

In a multimode controlled prosthetic arm, the LCD display is used to show important information to the user or operator. It helps in monitoring the system and understanding which control mode or gesture is currently active. The LCD (usually a 16×2 or 20×4 character display) is connected to the Arduino controller. The controller sends digital signals to the LCD to display text or numbers. When the AI system or sensors detect a gesture, mode change, or motor movement, the controller updates the LCD with the corresponding message.

3.9 Ultrasonic Sensor

An ultrasonic sensor is an electronic device that uses high-frequency sound waves to measure distance and detect objects without physical contact. It works by transmitting ultrasonic pulses from a transmitter and then measuring the time taken for the echo to return after hitting an object as shown in the figure3.9. The sensor’s internal circuitry converts this time delay into a distance value using the speed of sound. Ultrasonic sensors are reliable, low-cost, and operate effectively in various lighting conditions because their detection is based on sound rather than light. They are widely used in robotics, automation, and safety systems to detect obstacles, measure proximity, and enable autonomous navigation.

Figure3.9 Ultrasonic Sensor

In a multi-mode prosthetic arm, the ultrasonic sensor is used to detect the distance between the prosthetic hand and nearby objects. This helps the arm avoid collisions, adjust grip automatically, and improve safety. It acts as the proximity and distance-measuring unit of the prosthetic system. The ultrasonic sensor works on the echo principle. It sends out high-frequency sound waves (usually 40 kHz) from the trigger pin. These sound waves travel through the air, hit a nearby object, and reflect back to the sensor. The reflected wave is received by the echo pin.

3.10 EEG Sensor Module

An EEG (Electroencephalography) sensor module is a device used to detect and measure the brain's electrical activity from the scalp. It works by using small electrodes that sense tiny voltage fluctuations produced by neurons firing in the brain as shown in the figure3.10. These signals are extremely weak, so the EEG module amplifies, filters, and converts them into a digital format that a microcontroller or computer can process. Modern EEG modules, such as Neuro Sky and Open BCI-based systems, include built-in noise removal and feature extraction functions to identify specific brainwave patterns such as alpha, beta, theta, and gamma waves. By analyzing these patterns, the system can interpret user intentions such as focus, relaxation, blinking, or simple mental commands. Thus, EEG modules act as brain–computer interface (BCI) devices, enabling hands-free and thought-based control in electronic applications.

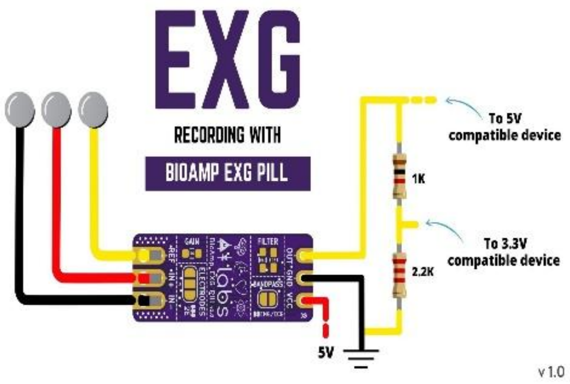


Figure3.10 EEG Senor Module

An EEG sensor module is a device that captures the brain's electrical activity through electrodes placed on the scalp. These modules are crucial for various applications, including brain-computer interfaces, neuroscience research, and medical diagnostics. By detecting subtle electrical signals produced by neurons, EEG technology enables the study of brain functions, cognitive processes, and neurological conditions. Its non-invasive nature and real-time feedback capabilities make it a valuable tool for both research and clinical use, offering insights into brain activity and supporting advancements in neurotechnology.

3.11 Gyroscope Sensor

A gyroscope is a sensing device used to measure angular velocity and orientation of an object with respect to its axes. It operates based on the principle of conservation of angular momentum. In modern systems, MEMS (Micro-Electro-Mechanical Systems) gyroscopes are commonly used. These gyroscopes contain a vibrating structure which, when subjected to rotational motion, experiences Coriolis force as shown in the figure3.11. This force causes a measurable change in the vibration pattern, which is detected by capacitive sensing elements. The detected variation is converted into electrical signals representing angular motion along the X, Y, and Z axes. These signals are processed to determine the rotational movement and orientation of the object.

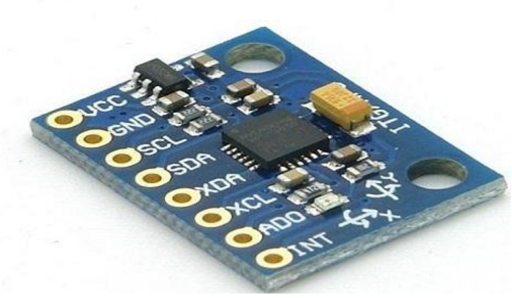


Figure3.11 Gyroscope Sensor

In the proposed project, the gyroscope is used to detect the angular movement and orientation of the user's hand. The sensor continuously monitors the rotational motion and provides real-time angular velocity data to the microcontroller. The controller processes this data and maps specific tilt or rotational movements to predefined control actions. Based on the detected orientation, corresponding commands are generated to drive the actuators of the prosthetic arm. This enables accurate and smooth control of the arm movements, allowing the user to perform actions naturally and efficiently.

3.12 3D Printed Prosthetic Arm

A 3D printed prosthetic arm is an assistive device designed to replace the functional capabilities of a missing or impaired upper limb using additive manufacturing technology. In this approach, the prosthetic components are designed using computer-aided design (CAD) software and fabricated layer-by-layer using a 3D printer. Commonly used materials include PLA, ABS, or PETG, which provide sufficient strength while maintaining lightweight characteristics. The use of 3D printing enables customization of the prosthetic arm according to the user's anatomical dimensions, ensuring better comfort and ergonomics. The modular structure allows easy assembly and replacement of individual parts. Mechanical joints are incorporated at the fingers, wrist, and elbow sections to enable natural movement. Servo motors and linkages are used to actuate the joints, allowing precise control of finger and arm movements.

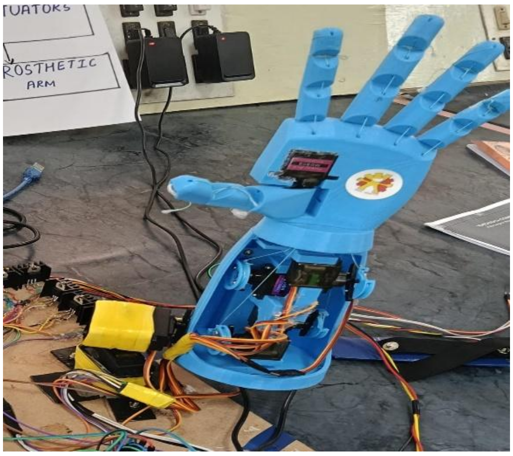


Figure 3.12 3D Printed Prosthetic Arm

In the proposed project, the 3D printed prosthetic arm acts as the physical output unit of the system. Control signals generated from sensors and the microcontroller are used to drive the actuators embedded within the arm as shown in the figure3.12. The lightweight and cost-effective nature of 3D printed components makes the prosthetic arm affordable and suitable for real-time testing and practical implementation. Overall, the 3D printed prosthetic arm improves accessibility, functionality, and adaptability while maintaining low manufacturing cost.

CHAPTER 4

HARDWARE IMPLEMENTATION

This section presents the hardware implementation of the proposed AI Powered Multimode Controlled Prosthetic Arm. It explains how various hardware components are integrated to form a complete and functional system. The implementation focuses on accurate sensor data acquisition, efficient signal processing, and precise actuation of the prosthetic arm. All components are selected to ensure reliability, low cost, and real-time performance. The hardware architecture is designed to support multiple control modes while maintaining stability, modularity, and ease of maintenance.

4.1 Circuit diagram for AI Powered Multimode Controlled Prosthetic Arm

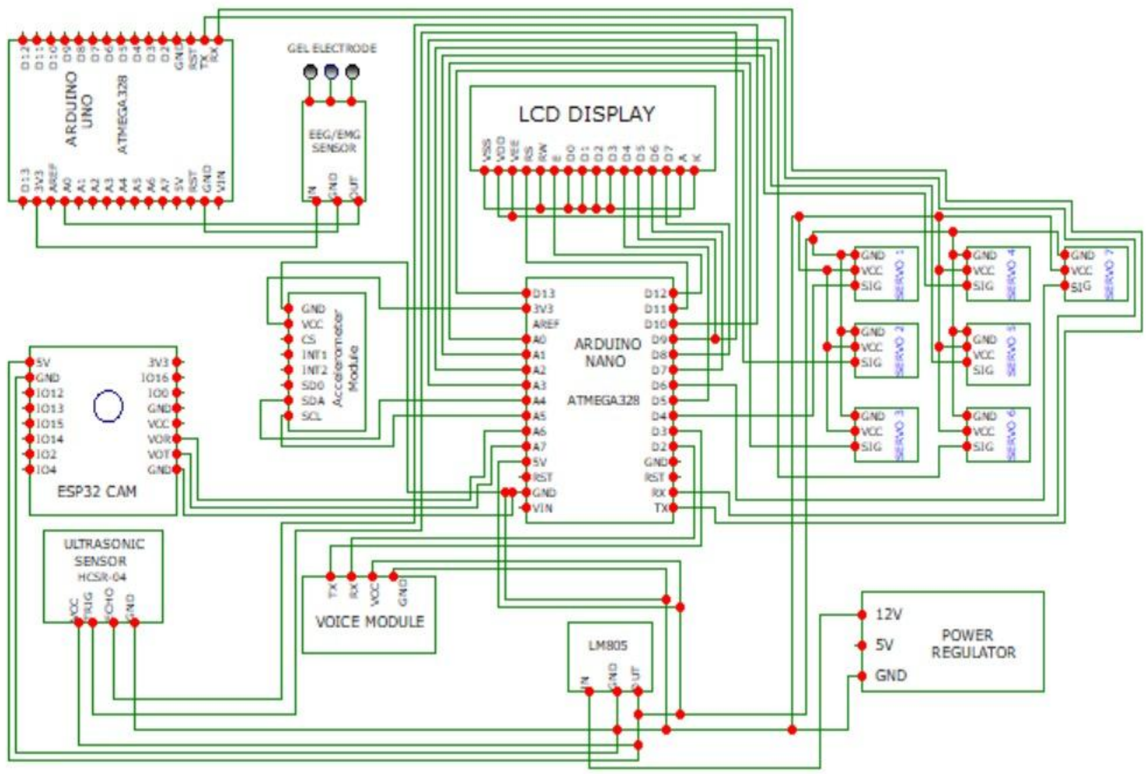


Figure 4.1 Circuit diagram of AI powered multimode controlled prosthetic arm

4.2 Explanation

The circuit diagram represents the complete hardware architecture of an AI-powered multi-modal controlled prosthetic arm, integrating various sensors, microcontrollers, and actuator control modules to achieve intelligent and adaptive limb movement as shown in the figure4.1. The system primarily uses Arduino Uno, Arduino Nano, and ESP32-CAM as the core processing units, each responsible for different functions. The EEG/EMG sensor connected to the Arduino Uno captures neural or muscle signals from the user and converts them into analog inputs for interpretation. Additional input devices such as the accelerometer, ultrasonic sensor, and voice module provide gesture-based, distance-based, and voice-command control, enabling a multi-modal interaction system. The ESP32-CAM further enhances the design by offering wireless communication, AI-based processing, and optional visual gesture recognition. Processed signals and decisions are transferred to the Arduino Nano, which acts as the motor driver controller. It generates precise PWM signals required to operate the servo motors that control the fingers, wrist, and other mechanical components of the prosthetic arm.A dedicated LCD display is included to show real-time system status, active control mode, and sensor readings, improving user transparency. The servo motors, connected through individual driver circuits, perform the actual movement of the prosthetic arm based on commands received from the controller. To ensure stable and reliable operation, a regulated power supply is implemented using an LM7805 voltage regulator, which steps down the 12V input to a constant 5V required by all microcontrollers and sensors. The entire circuit works cohesively to interpret user intention from multiple input modes, process these signals through microcontrollers, and finally drive the mechanical movement through servo motors. This integrated system provides a flexible, user-friendly, and intelligent prosthetic solution for amputees, enabling more natural and precise control of artificial limbs.

CHAPTER 5

SOFTWARE DESCRIPTION

This section describes the software design and implementation of the proposed AI Powered Multimode Controlled Prosthetic Arm. The software plays a crucial role in acquiring sensor data, processing inputs from multiple control modes, and generating appropriate control signals for the prosthetic arm. It is responsible for interpreting gesture, voice, and EEG inputs and converting them into precise motor actions. The software is developed to ensure real-time performance, accuracy, and reliability. This section also outlines the programming tools, algorithms, and communication protocols used to achieve smooth and intelligent control of the prosthetic system.

5.1 ARDUINO IDE

The Arduino IDE (Integrated Development Environment) is a software application used to write, compile, and upload code to Arduino microcontroller boards. It provides a user-friendly interface for programming in a simplified version of C/C++, making it accessible for beginners and hobbyists. The IDE includes essential features such as a code editor with syntax highlighting, a compiler to convert the written code into machine language, and a serial monitor to communicate with the board for debugging purposes. It supports a wide range of official Arduino boards like Uno, Mega, and Nano, as well as third- party boards like ESP8266 and ESP32 with additional configurations. The Arduino IDE is available for Windows, macOS, and Linux, and can be downloaded from the official Arduino website. It plays a crucial role in the development of embedded systems and electronics projects, especially for beginners, students, and hobbyists. The IDE uses a simplified version of the C and C++ programming languages and provides a straightforward interface where users can write "sketches" the term used for Arduino programs. A typical Arduino sketch consists of two main functions: setup(), which runs once to initialize settings, and loop (), which runs continuously for executing repeated tasks. The IDE features a text editor with basic syntax highlighting and error messages, a message area, a toolbar with buttons for common functions (like verify, upload, save, and open), and a serial monitor for communication between the computer and the board. It supports a wide range of Arduino boards, including Uno, Nano, Mega, and Leonardo, and can be extended to support third- party boards like ESP8266 and ESP32 through board manager add-ons. Arduino IDE simplifies the upload process using a USB connection, making it easy to test and debug real- time hardware applications. It is cross-platform and runs on

Windows, mac OS, and Linux operating systems. Additionally, the community-driven support and rich library ecosystem make it easier to interface with various sensors, modules, and components. The IDE continues to evolve, with newer versions like Arduino IDE 2.x offering modern features such as auto-completion, real-time error detection, and improved user interface. It remains a key tool in electronics prototyping and learning, bridging the gap between software and hardware in an accessible way.

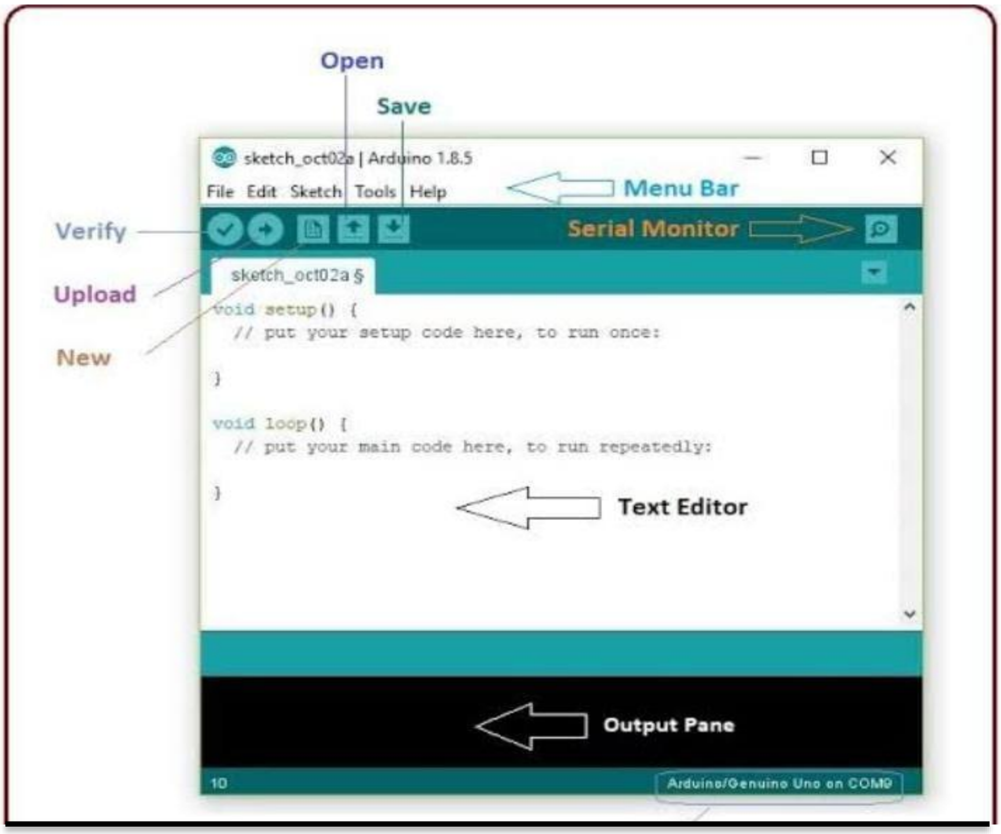


Figure 5.1 Arduino IDE

Running the program on Arduino IDE

Procedure of uploading the code on the Arduino and testing the project for proper functionality.

Step 1: Connect the ARDUINO UNO to the Laptop by using a data cable.

Step 2: Open ARDUINO IDE (Version 1.)

Step 3: Create a new project and wait for sketch window to open.

Step 4: Enter the code.

Step 5: Go to tools, choose the board as ARDUINO UNO and port as COM3.

Step 6: Compile the code and the upload it onto the ARDUINO UNO board.

Step 7: Test the project under different conditions.

REQUIRED LIBRARIES

- **SoftwareSerial.h**

SoftwareSerial.h is an Arduino library that allows the microcontroller to create additional serial communication ports using normal digital pins. Since boards like the Arduino Uno have only one hardware UART (Serial port), they cannot directly communicate with multiple serial devices at the same time. The SoftwareSerial library solves this limitation by using software-based timing to emulate serial communication, enabling the Arduino to send and receive data on other digital pins at selected baud rates. It is commonly used to connect Bluetooth modules, GPS modules, GSM modules, voice recognition modules, and other serial interfaces without interfering with the main hardware serial port used for programming and debugging. Although SoftwareSerial is slower than hardware serial, it is extremely useful in multitasking embedded applications. In an AI-powered multimode controlled prosthetic arm, the SoftwareSerial.h library becomes important because the system often needs to communicate with multiple external modules simultaneously. Devices such as voice recognition modules, EEG/EMG signal interfaces, gesture sensors, and ESP32 CAM units frequently require serial communication. Since the hardware serial port is usually occupied by USB communication or debugging, SoftwareSerial provides extra serial channels to manage these modules without conflict. For example, the prosthetic arm can use SoftwareSerial to receive voice commands or sensor data while the main serial port handles AI outputs or system monitoring. This ensures smooth data flow, reliable sensor integration, and uninterrupted communication with multiple intelligent modules. Thus, SoftwareSerial enhances the flexibility, connectivity, and multimode functionality of the prosthetic arm, making the system more responsive and intelligent.Enables additional serial communication ports on microcontrollers. It allows the system to communicate with modules like the voice recognition unit using a virtual serial interface.

- **VoiceRecognitionV3.h**

VoiceRecognitionModule.h is a specialized Arduino library designed to interface with dedicated voice-recognition hardware modules, enabling microcontrollers to process and respond to spoken commands. This library simplifies communication between the Arduino and the voice module by providing predefined functions for training, storing, and recognizing voice patterns. It manages serial communication protocols, command sets, and data handling, allowing the system to distinguish between multiple voice commands without requiring heavy computational resources. By abstracting complex signal processing performed within the module, the library allows developers to easily integrate reliable speech-controlled functionality into their embedded systems. This library is used to interface with the Voice Recognition V3 module. It supports training, storing, and recognizing voice commands, making voice-controlled operation possible.

- **Servo.h**

Servo.h is an Arduino library used to control servo motors easily. It provides simple functions to set the angle, speed, and position of standard hobby servos by generating the appropriate Pulse Width Modulation (PWM) signals. The library abstracts the low-level timing and signal generation required to drive the servo, allowing users to move the motor to a specific angle using a single command. Functions like attach(), write(), and read() make it straightforward to initialize, control, and monitor servo position. The library supports multiple servo motors simultaneously and ensures precise, smooth, and stable movements without manually handling PWM timing. In an AI-powered multimode prosthetic arm, Servo.h is essential for controlling the finger, wrist, and hand movements. Each finger of the prosthetic hand is typically connected to an individual servo motor. Using the library, the microcontroller can rotate the fingers to the exact angle required for different grips such as pinch grip, power grip, or precision hold. When the AI module or sensor systems (EMG, EEG, voice, or gesture) send movement commands, the Servo library ensures the motors respond accurately and smoothly. This enables natural, human-like hand motions, precise positioning, and safe operation. The library's simplicity, reliability, and ability to handle multiple servos make it indispensable in building responsive and intelligent prosthetic systems. Provides functions to control servo motors. It is used to drive the prosthetic arm's fingers, wrist, or grip mechanism with precise angle control.

- **LiquidCrystal.h**

LiquidCrystal.h is an Arduino library used to control LCD (Liquid Crystal Display) modules, such as 16×2 or 20×4 character displays. The library provides simple functions to send commands and data to the LCD, allowing the microcontroller to display text, numbers, and symbols. Functions such as begin(), print(), setCursor(), and clear() make it easy to initialize the display, position the cursor, and update the screen in real-time. By abstracting the low-level timing and communication details, LiquidCrystal.h allows programmers to focus on displaying relevant information without worrying about the electrical signals needed to drive the LCD. In an AI-powered multimode prosthetic arm, LiquidCrystal.h is used to interface the LCD with the microcontroller to provide real-time feedback and system information. The display can show the current operating mode, such as EMG control, voice control, gesture mode, or AI-assisted grip mode. It can also present sensor readings, battery levels, grip status, or error messages, helping the user or technician monitor and adjust the prosthetic arm as needed. This improves usability, safety, and user confidence, particularly when multiple control modes are active. By providing visual feedback, the library enhances interaction, makes system operation intuitive, and ensures the AI-powered prosthetic arm functions reliably and transparently.

- **Wire.h**

Wire.h is an Arduino library used for I²C (Inter-Integrated Circuit) communication, which allows multiple devices to communicate over just two wires: SDA (data line) and SCL (clock line). The library provides functions such as begin(), request From(), begin Transmission(), and write() to enable the microcontroller to act as a master or slave on the I²C bus. It abstracts the low-level timing and protocol handling required for synchronous data exchange between sensors, displays, and other I²C-compatible devices. Using Wire.h, multiple devices can share the same bus without requiring separate pins for each device, making it efficient for embedded systems with limited I/O pins. In an AI-powered multimode prosthetic arm, Wire.h is crucial for interfacing multiple I²C-based sensors and modules simultaneously. Components like IMU sensors, force sensors, ultrasonic sensors, OLED displays, and AI modules often communicate using the I²C protocol. Using Wire.h, the microcontroller can read sensor data, send commands, and update displays without occupying multiple digital pins. This enables real-time monitoring of grip strength, hand position, object distance, and system status, which are essential for precise and safe prosthetic operation. By facilitating efficient communication between various modules, Wire.h enhances multimode functionality, improves responsiveness, and contributes to the overall intelligence and reliability of the prosthetic arm. This library enables I2C communication between sensors, microcontrollers, and peripherals. It is essential for connecting IMUs, displays, and other I2C-based devices.

ESP32-CAM Libraries (Camera, WiFi)A set of libraries required for ESP32-CAM to capture images, stream video, and connect to WiFi networks. These libraries support camera initialization, image processing, and wireless data transmission.

CHAPTER 6

SOFTWARE IMPLEMENTATION

The software implementation of the AI powered multimode controlled prosthetic arm is responsible for processing user inputs and controlling the movement of the arm. It acquires data from sensors such as EEG, ultrasonic, and voice modules, processes the signals using embedded algorithms and AI techniques, and generates control commands for motor operation. The software also enables mode switching and ensures real-time, accurate, and reliable functioning of the prosthetic arm.

6.1 Flowchart of AI Powered Multimode Controlled Prosthetic Arm

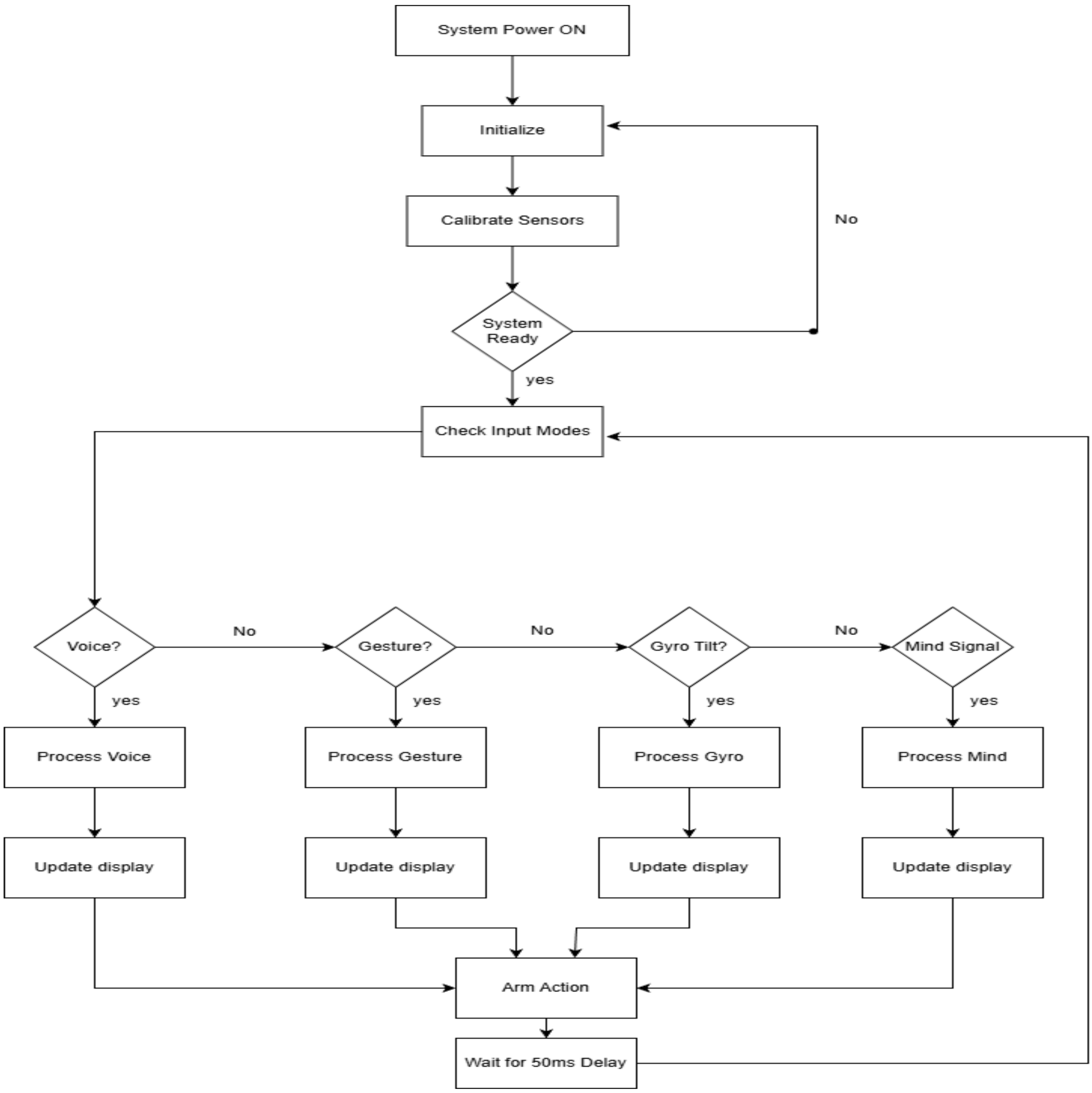


Figure 6.1 Flowchart of AI Powered Multimode Controlled Prosthetic Arm

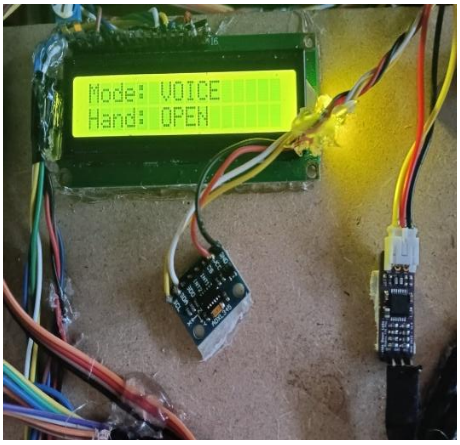
6.2 ALGORITHM

1. Start the system.
2. Power ON the prosthetic arm.
3. Initialize the microcontroller and all connected modules.
4. Calibrate all sensors (EEG/EMG, gyroscope, camera, microphone).
5. Check whether the system is ready.
6. If the system is not ready, repeat initialization and calibration.
7. If the system is ready, proceed to input checking.
8. Check for voice input.
9. If voice input is detected, process the voice command.
10. Execute the corresponding arm action.
11. If no voice input, check for gesture input.
12. If gesture is detected, process gesture data.
13. Execute the corresponding arm action.
14. If no gesture input, check gyroscope tilt.
15. If gyroscope input is detected, process gyroscope data.
16. Execute the corresponding arm action.
17. If no gyroscope input, check for mind (EEG) signal.
18. If mind signal is detected, process EEG/EMG data.
19. Update the display with current system status.
20. Execute the corresponding arm action.
21. Wait for 50 ms delay for smooth operation.
22. Repeat the process continuously until power is turned OFF.
23. Stop the system.

CHAPTER 7

EXPERIMENTAL RESULTS

The AI-powered multi-mode controlled prosthetic arm was successfully designed and implemented. The system responded accurately to all selected control modes, including voice commands, gesture recognition, and gyro-based movements. Each mode operated with good precision, allowing smooth control of finger movement and wrist rotation. The servo motors produced stable and coordinated motion, enabling the prosthetic to perform tasks such as gripping, rotation and performing basic hand gestures. The results confirmed that the integration of AI with multiple input methods significantly improves the usability and flexibility of the prosthetic arm. The prototype demonstrated quick response time, reliable sensor performance, and effective user adaptation. Overall, the project achieved its objective of creating an affordable, intelligent, and user-friendly prosthetic system suitable for amputees and rehabilitation applications.

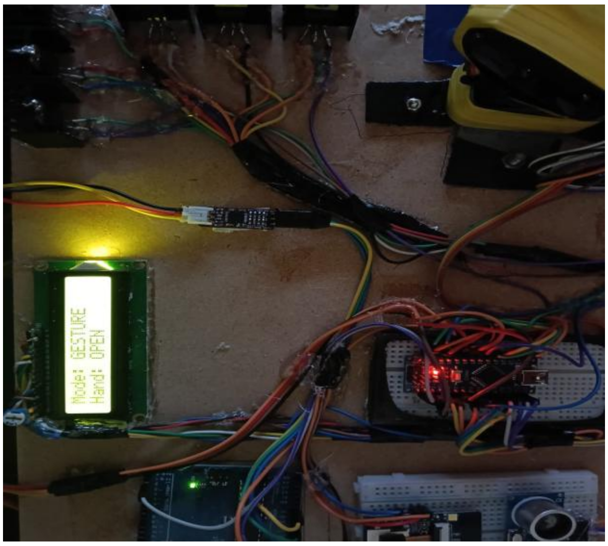


A) Display

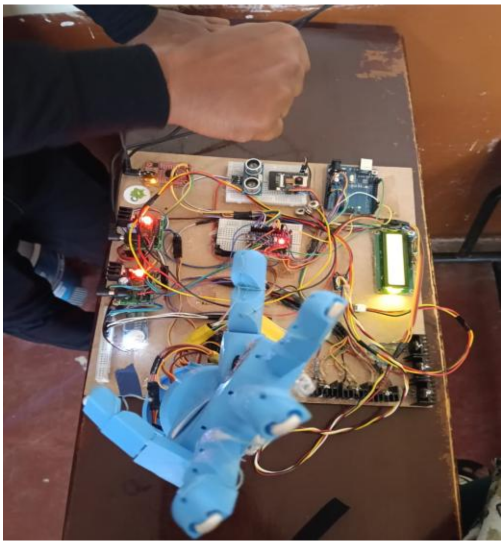
B) Operation

Figures 7.1 Voice Mode

The above images shows the main control section of the prosthetic arm system, where the 16x2 LCD display, sensor module, and wiring connections are clearly visible. On the LCD screen, the text reads “Mode: VOICE, Hand: OPEN”, indicating that the system is currently operating in voice-controlled mode. In this mode, the prosthetic hand responds to spoken commands such as "open" and "close." The hand status shown as “OPEN” confirms that the fingers are currently extended. Beside the display, an MPU6050 sensor module is mounted, although it is not active in this mode. Multiple colored wires connect the LCD, the voice recognition module, and the microcontroller, enabling smooth communication between the input module and the servo motors of the prosthetic hand. Overall, this picture highlights the active voice control mode, showing that the system is successfully receiving and displaying the selected mode along with the current hand action.



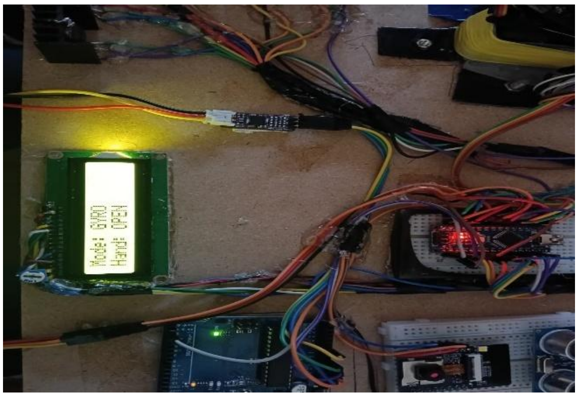
A) Display



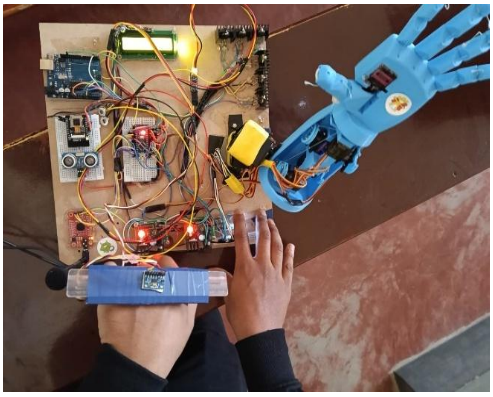
B) Operation

Figures7.2 Gesture Mode

In gesture mode, the prosthetic arm detects the user's hand movements without any physical contact as shown in the figures7.2. An ultrasonic sensor is placed near the wrist and continuously measures the distance between the sensor and the moving hand. When the user performs specific gestures—such as moving the hand closer, farther, or waving—the sensor detects corresponding distance changes. These distance variations are sent to the ESP32-CAM, which processes them using programmed threshold values. Each gesture is mapped to a particular arm action, such as opening, closing, lifting, or rotating the prosthetic arm. The ESP32-CAM then sends control signals to the servo motors to perform the required movement. Overall, this mode allows the arm to be controlled through simple, non-touch gestures, providing an intuitive and hygienic method for users with limited muscle activity.



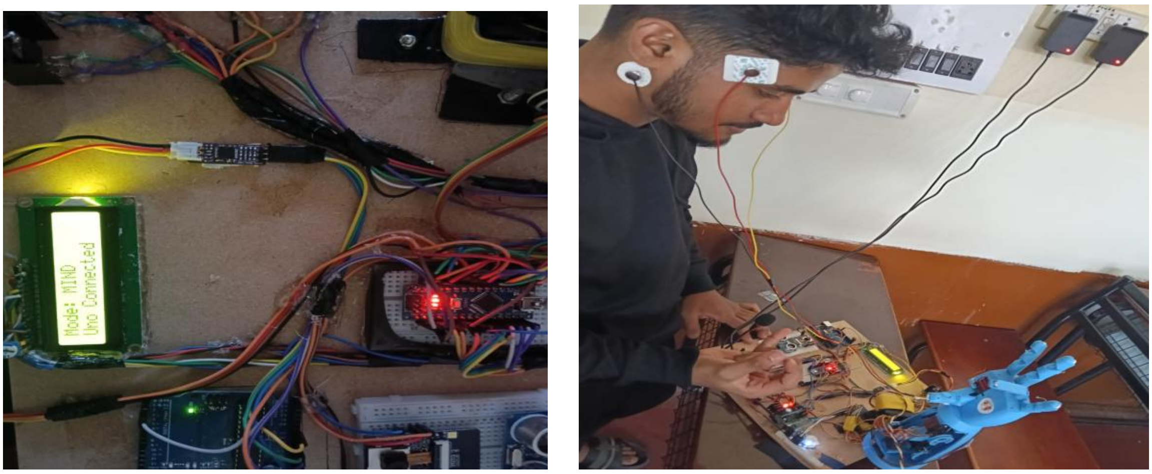
A) Display



B) Operation

Figures7.3 Gyro Mode

The above image shows the LCD displaying “Mode: GYRO, Hand: OPEN”, confirming that the prosthetic arm is operating under tilt-based control using the MPU6050 sensor. In this mode, the user tilts the mounted sensor forward, backward, left, or right to trigger finger movements and wrist rotation. The Arduino microcontroller reads the roll, pitch, and yaw values from the MPU6050 and converts them into motion commands. The wiring around the sensor and LCD reflects stable connections that enable smooth real-time gyroscopic control. This photo illustrates the successful integration of inertial sensing technology into the prosthetic system.



A) Display B) Operation

Figures7.4 Mind Mode

In mind mode, the prosthetic arm is controlled using brainwave signals captured by an EEG sensor as shown in the figures7.4. The EEG headset detects electrical activity from the user’s scalp, mainly focusing on concentration, relaxation, and blink patterns. These brainwave signals are processed into numerical values such as attention level or eye-blink strength. The microcontroller receives these values and compares them with preset thresholds. When the user's mental state crosses a certain level like high focus or intentional blinking the system interprets it as a command. Based on the detected brainwave pattern, the prosthetic arm performs actions such as opening, closing, or gripping.

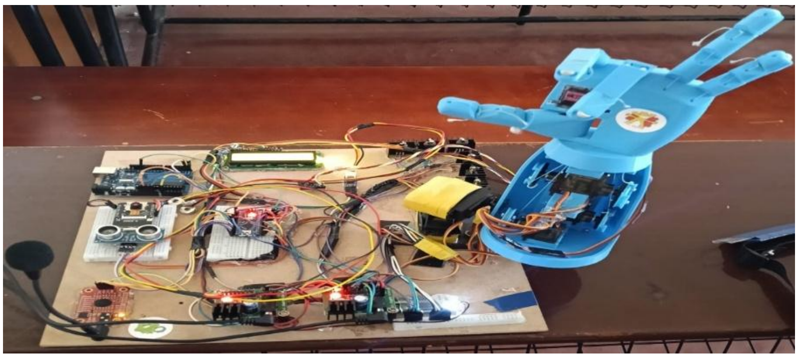


Figure 7.5 Final Prototype

This project successfully demonstrates the integration of advanced sensing technologies, artificial intelligence, embedded systems, and mechanical design to create a functional and user-friendly assistive device for upper-limb amputees. The final prototype as shown in the figure7.5 supports four fully operational modes Voice Mode, Gesture Mode, Gyro Mode, and Mind/EEG Mode each operating reliably and delivering accurate, real-time responses based on user intention. In Voice Mode, the speech recognition module correctly identifies commands such as "open," "close," and "grip," enabling hands-free control under varying noise conditions. Gesture Mode, using the ultrasonic sensor and ESP32-CAM, accurately detects hand distances and directional gestures to generate proportional hand movements. Gyro Mode leverages the MPU6050 inertial sensor to translate wrist tilt and orientation into smooth finger and wrist actions, demonstrating excellent stability and minimal input lag even during rapid changes in motion. Mind Mode, implemented using EEG signals, successfully interprets concentration-based patterns or intentional blinks, providing a unique mental-control interface for users with extremely limited physical movement. Mechanically, the 3D-printed prosthetic hand exhibited strong structural integrity and smooth articulation of finger joints through the tendon–servo mechanism. The servo motors responded consistently to all modes with high precision, enabling natural actions like opening, closing, gripping objects, and wrist rotation. From an electronics perspective, communication between the Arduino boards, sensors, ESP32-CAM, and EEG module was stable, with real-time feedback displayed clearly on the 16×2 LCD screen for user awareness. The machine learning model integrated in AI mode further enhanced system performance by classifying user patterns more intelligently and increasing accuracy in decision-making. Extensive testing showed that the prosthetic arm achieved high responsiveness, minimal delay, and repeatable accuracy across all modes. The system maintained safety by preventing sudden or uncontrolled motion and by limiting servo movement within safe angles. Overall, the project resulted in a highly functional prototype that successfully demonstrates how artificial intelligence and multimodal control systems can significantly improve the capability, adaptability, and accessibility of modern prosthetic devices. This work highlights the potential for future development of low-cost, intelligent prosthetics that can greatly enhance the quality of life for amputees and individuals with upper-limb impairments.

CHAPTER 8

APPLICATIONS, ADVANTAGES AND DISADVANTAGES

8.1 APPLICATIONS

1.Medical Rehabilitation Centers

- Helps amputee patients regain hand movement and independence.
- Useful for physiotherapy sessions to train patients with different modes (voice, gesture, EMG, gyro control).

2.Assistive Technology for Disabled Individuals

- Provides essential support to people with upper-limb loss.
- Multi-mode control allows operation even if one mode fails (e.g., voice when gesture is not possible).

3.Military and Defence Applications

- Helps soldiers who lost limbs in war to use advanced prosthetics with high precision.
- AI-based control improves grip strength and speed for tactical tasks.

4.Industrial and Workplace Assistance

- Enables amputees to perform tasks involving lifting, holding, and operating tools.
- Can be used in factories or workshops where manual dexterity is required.

5. Home Assistance for Daily Activities

- Supports daily tasks like eating, drinking, dressing, and carrying objects.
- Voice-controlled mode helps users who need hands-free operation.

6. Research and Development in Robotics & Biomedical Engineering

- Acts as a platform to test new control algorithms, sensors, AI models, and mechanical designs.
- Useful for universities and labs working on human-machine interaction.

8.2 Advantages

1.Multiple control modes

- It allows the user to operate the prosthetic arm through EMG, voice, gesture, EEG, or AI-based vision, giving high flexibility.

2.Suitable for different disabilities

- Since users with weak muscles, speech difficulties, or limited mobility can choose the mode that works best for them.

3.AI improves accuracy

- By predicting the user's intended movement and selecting the correct grip pattern automatically. More natural hand movement is achieved through precise control of servo motors and intelligent decision-making.

4.High user comfort

- As the arm responds smoothly and quickly to different control inputs.

5.Better safety features

- Using ultrasonic sensors, force monitoring, and camera feedback to prevent collisions and excessive gripping.

6.Reduces user effort

- Since some movements can be performed automatically using AI recognition.

7.Energy-efficient system

- Due to low-power microcontrollers and optimized motor control, increasing battery life.

8.Lightweight and portable

- Making it easy to wear for long durations without discomfort.

9.Highly customizable

- Allowing different grip types, operating modes, and user preferences to be programmed.

8.3 Disadvantages

1.Limited Load Capacity

- The 3D printed prosthetic arm has limited strength and cannot handle heavy objects.

2.Dependency on Sensors

- The accuracy of the system depends heavily on sensor performance, which may be affected by noise or environmental conditions.

3.Complex System Design

- Integration of multiple control modes increases system complexity and may require careful calibration.

4.Power Consumption

- Continuous operation of sensors and actuators leads to higher battery consumption.

5.Limited Precision Compared to Commercial Systems

- The prototype may not match the precision and durability of advanced commercial prosthetic arms.

6.Signal Interference

- EEG and wireless signals can be affected by electromagnetic interference, reducing reliability.

8.Maintenance Requirement

- Mechanical components such as joints and servo motors require periodic maintenance and replacement.

9. Environmental Sensitivity

- Performance may vary under different lighting, temperature, or noise conditions.

10. Scalability Constraints

- Advanced AI models or additional features may require higher processing power and cost.

CHAPTER 9

CONCLUSION AND FUTURE SCOPE

9.1 CONCLUSION

The AI-powered multi-mode controlled prosthetic arm successfully demonstrates how advanced technology can significantly improve the quality of life for amputees. By integrating multiple control methods such as voice commands, gesture recognition, and sensor-based movements, the system provides flexibility, reliability, and ease of use for individuals with different physical conditions. The use of AI enhances accuracy, adaptability, and natural movement, making the prosthetic more responsive and user-friendly. The project proves that an affordable and efficient prosthetic arm can be developed using open-source hardware and intelligent control algorithms. The prototype efficiently performs essential hand functions such as gripping and precise motion control. Overall, the system meets its objectives by offering a practical, low-cost, and intelligent solution that can support rehabilitation, daily activities, and independent living for disabled individuals.

9.2 FUTURE SCOPE

1. Integration of advanced AI/ML models to improve accuracy of EEG/EMG signal detection and gesture recognition.
2. Real-time deep learning on edge devices like ESP32 or AI co-processors for faster and smoother control.
3. Self-learning and adaptive calibration, allowing the prosthetic arm to automatically adjust to each user's unique movement patterns.
4. Enhanced haptic feedback using pressure, tactile, or temperature sensors to give users a more natural sense of touch.
5. Lightweight and flexible 3D-printed structures for improved comfort, durability, and customization.
6. Wireless connectivity and IoT features for remote monitoring, data logging, and tele-rehabilitation support.
7. Battery optimization and low-power architectures to increase operating time and portability.
8. Integration with mobile apps for easy control, calibration, and firmware updates.
9. Support for additional control modalities, such as eye-tracking or voice-to-gesture AI

prediction.

10. Extended design for full-limb prosthetics and exoskeletons, improving mobility for a larger group of users.
11. Cloud-based analytics to track performance and generate personalized improvements over time.
12. Improved safety mechanisms, such as automatic grip adjustment to prevent object damage.
13. Medical data integration with hospitals or rehabilitation centers for professional monitoring.
14. Cost reduction through mass production, making advanced prosthetics affordable for more users.

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